Assignment 3

Mathematics and Tables Using LaTeX

Que 1

Write LaTeX code to produce the following expression:

$$\overbrace{a+b+c}^{\text{First Group}}+\underbrace{x+y+z}_{\text{Second Group}}=\overbrace{a+\underbrace{b+c}_{\text{Sum 1}}}^{\text{Total}}$$

Que 2

Create the following array using the array environment:

$$A=\left[\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{array}\right]$$

Que 3

Write the following piecewise function using the cases environment:

$$f(x)=\begin{cases} x^2+2x+1, & x<0, \\ \sqrt{x+1}, & 0\leq x<4, \\ \dfrac{x^2-1}{x-1}, & x\geq 4. \end{cases}$$

Que 4

Typeset the following matrix and label its different sections using overbrace:

$$\begin{bmatrix} \overbrace{a_{11} & a_{12} & a_{13}}^{\text{Column Group 1}} & \overbrace{a_{14} & a_{15}}^{\text{Column Group 2}} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{bmatrix}$$

Que 5

Create the following expression:

$$F(x)=\underbrace{\sum_{i=1}^{n}\left(a_i x_i^2+b_i x_i+c_i\right)}_{\text{Quadratic Contribution}}+\underbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{\text{Matrix Contribution}}$$

$$\underbrace{F(x)}_{\text{Total Energy}}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Que 6

Create the following table with proper alignment, borders, and headings.

Roll No.	Name	Department	Semester	CGPA
101	Amit	CSE	2	8.45
102	Neha	CSE	2	9.12
103	Rahul	ECE	4	8.76
104	Priya	ECE	4	9.01

Que 7

Create the following table using multicolumn.

Academic Performance		
Student	Midterm	Final
A	78	89
B	85	92
C	91	94

Que 8

Create the following table using multirow.

Department	Course	Credits	Grade
CSE	Data Structures	4	A
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Que 9

Create the following scientific table using xcolor and colortbl.

Model	Parameters	FLOPs	Accuracy	Energy (kWh)
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

Que 10

Create a student-performance table with conditional cell coloring.

Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98

Color Rules:

- Marks ≥ 90: Green
- Marks 75–89: Yellow
- Marks < 75: Red

Que 11

Create a table containing paragraphs inside cells.

Technique	**Description**	**Advantages**	**Limitations**
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

Que 12

Create a professional table using the booktabs package.

Table 1: Comparison of Different Models

Model	**Parameters**	**FLOPs**	**Accuracy**	**Energy**
ResNet18	11.7M	1.82G	94.21	0.00093
MobileNetV2	3.5M	0.30G	92.84	0.00061
EfficientNet-B0	5.3M	0.39G	93.52	0.00072

Table 1 presents a comparison of different models.

Que 13

Create a table where one cell contains another table.

Model	**Architecture**	**Performance**
CNN	Conv → ReLU → Pool	Metric: Accuracy 94%, F1 Score 93%
ResNet	Conv → BN → ReLU	Metric: Accuracy 96%, F1 Score 95%
ViT	Attention → MLP	Metric: Accuracy 97%, F1 Score 96%

Que 14

Create the following table containing mathematical expressions.

Model	**Loss Function**	**Parameters**	**Complexity**
Linear	$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	10K	$O(n)$
CNN	$L = -\sum_i y_i \log(\hat{y})$	1.2M	$O(n^2)$
Transformer	$L = -\sum_i y_i \log p_i$	110M	$O(n^2 d)$

— End of Assignment —